

On the Invariant Theory of Forms in n Variables

By Miss *Emmy Noether* in Erlangen.

Journal für die reine und angewandte Mathematik 139 (1911), pp. 118–154

Source-audited working draft containing §6 only; original printed pp. 137–141.

§ 6. Explicit Representation of the Invariant Formations. Folding and Differentiation Processes.

The results of the preceding sections allow us to complete the theorem on symbolic representability stated in § 2 (the first fundamental theorem) by adding explicit representability; namely, we show:

Theorem IV. All invariant formations of arbitrary ground forms can be represented explicitly by matrix products; that is, aside from variable rows p_ρ , only the rows $S^\sigma, \dots, T^\tau$ of the original forms, or the rows P, Q, R derived from them by substitution (9) or (11), enter the final expressions.¹

For the proof, suppose that an invariant formation depends, besides variable rows, on the rows $S^\sigma, \dots, T^\tau$, and suppose that these rows have been resolved far enough into individual rows, $S^\sigma = S^{\sigma_1} \dots S^{\sigma_i}, \dots, T^\tau = T^{\tau_1} \dots T^{\tau_k}$, to permit a representation by determinants. Let the rows be ordered in a sequence $S^\sigma, \dots, T^\tau$ that is arbitrary in itself but fixed once and for all. We select an aggregate of determinants that depends on the first total row $S^\sigma = S^{\sigma_1} \dots S^{\sigma_i}$, not merely on individual component rows. By §§ 3–5 this dependence is a consequence of the general identities. If we now resolve a variable row p_ρ into the rows y, v , or one of the later rows T into the individual rows t , respectively τ , then the most general identities are composed of (20) and (21). These identities always lead to an expression containing the row $S^\sigma = S^{\sigma_1} \dots S^{\sigma_i}$ explicitly, namely the left-hand side of (20) and (21). Indeed, (19) shows that only the complete sum on the right—not a part of it corresponding to a term of (20a)—has the property of depending only on S^σ .

As the procedure continues, every row that has once entered explicitly remains explicit; applying the identities can at most require several rows to be combined into one, that is, substitution (9). If only a single row T^τ remains to be brought to explicit form, two cases are possible. There may still be a free variable row, one not connected with other rows by substitution (9); resolving it into component rows y or v completes the procedure and gives, as in (31), polars of one or more forms containing all rows explicitly. If no free variable row remains, then, from the way they arose, we recognize in the totality of the expressions that contain the individual rows t or τ but depend on T^τ the terms of a decomposition identity. By § 5 these always admit explicit representation in all rows after substitution (11). This proves the theorem in full generality. It should also be noted that when only two symbol rows occur, substitution (11) never appears. Since $\sigma, n - \sigma, \tau, n - \tau < n$, no variable rows can enter only when the two symbol rows are united in a single determinant and thus occur explicitly; as soon as variable rows enter, however, (34) and (33) show that substitution (11) becomes unnecessary.

It follows that, in order to generate all invariant formations, it suffices to unite the symbol rows, after adjoining variable rows, into all possible matrix products. The most general matrix product is of the form

$$(P^{\mu_1} \dots P^{\mu_i} \mid Q_{\nu_1} \dots Q_{\nu_k}), \quad \sum \mu = \sum \nu, \quad (40)$$

where the P and Q may be rows of the original forms, may have arisen from the original rows through a sequence of substitutions (9) or (11), or may be variable rows. The expressions (40) can be generated by successively uniting two symbol rows at a time into a matrix product, with the help of variable rows. Indeed, from

$$(q_{n-\mu-\lambda} P^\mu \mid Q_{\nu_1} p_{n-\nu_1-\lambda}) = (R_\lambda Q_{\nu_1} p_{n-\nu_1-\lambda})$$

one obtains, by uniting with the row Q_{ν_2} , an expression

$$([R_\lambda Q_{\nu_1}]^{n-\lambda-\nu_1} \mid Q_{\nu_2} p_{n-\nu_1-\nu_2-\lambda}) = (q_{n-\mu-\lambda} P^\mu \mid Q_{\nu_1} Q_{\nu_2} p_{n-\nu_1-\nu_2-\lambda}).$$

Continuing the procedure therefore yields an expression (40). If P and Q are not rows of the original forms, (9) and (11), together with what has just been shown, imply that they too arose by a sequence of unions of two symbol rows. Thus:

Theorem V. The process for generating all invariant formations—the folding process—consists in successively uniting two symbol rows at a time into a matrix product, with the help of variable rows.

¹The current R823 German source separates the compound *Schluß ausdrücke* at R823 line 4048. The original print has the single compound *Schlußausdrücke*. The printed compound is the coherent reading and is followed in translation. This source defect has been recorded for correction in the upstream German corpus.

From two symbol rows S^σ, T^τ , where $\sigma \geq \tau$, the following matrix products can be formed:

$$(q_{n-\sigma-\lambda} S^\sigma \mid T_{n-\tau} p_{\tau-\lambda}), \quad (\lambda = 1, 2, \dots, \min(\tau, n - \sigma)), \quad (41)$$

$$(q_{n-\tau-\mu} T^\tau \mid S_{n-\sigma} p_{\sigma-\mu}), \quad (\mu = \sigma - \tau + \lambda), \quad (42)$$

$$(q_{n-\tau-\nu} T^\tau \mid S_{n-\sigma} p_{\sigma-\nu}), \quad (\nu = 1, 2, \dots, \sigma - \tau). \quad (43)$$

From (31), however, the following values are obtained for (42) and (43):

$$(q_{n-\sigma-\lambda} S^\sigma \mid T_{n-\tau} p_{\tau-\lambda}) + \sum_{\alpha} \Pi(q_{n-\sigma-\lambda-\alpha} S^\sigma \mid T_{n-\tau} p_{\tau-\lambda-\alpha}), \quad (42a)$$

$$\Pi(q_{n-\sigma} S^\sigma)(T_{n-\tau} p_\tau) + \sum_{\beta} \Pi(q_{n-\sigma-\beta} S^\sigma \mid T_{n-\tau} p_{\tau-\beta}). \quad (43a)$$

Thus the forms (42) can be expressed linearly through (41) and polars of (41), and the forms (43) through polars of the ground form and of the forms (41). Likewise, (31) gives the linear dependence of the forms (41) on (42) and polars of (42). By Clebsch's definition,² the forms (42) are therefore equivalent to the forms (41), whereas (43) leads back to the ground form. The generating process is thus given equally well by (41) or (42). We shall define the process (41), which is simpler with respect to the number λ , as the folding process. The number λ , the defect of the folding (cf. § 2), gives the number of rows missing from the determinant product, namely in the matrix product that contains the maximum number of rows for the given folding. Since λ runs only up to the smaller of τ and $n - \sigma$, where $\sigma \geq \tau$, we have

$$\lambda \leq \frac{n}{2} \quad (n \text{ even}); \quad \lambda \leq \frac{n-1}{2} \quad (n \text{ odd}). \quad (44)$$

The reduction of (42) and (43) to (41) remains valid when further folding has turned the rows p, q into symbol rows; for by (36),

$$(A^{n-\tau-\varkappa} T^\tau \mid S_{n-\sigma} B_{\sigma-\varkappa}) = \sum_{\alpha=0, \varkappa-\sigma+\tau}^{\tau, n-\sigma} \varepsilon(R^{\tau-\alpha} B^{n-\sigma+\varkappa} \mid A_{\tau+\varkappa} R_{n-\sigma-\alpha}),$$

where the rows R arose from S and T according to (33). Hence these forms containing only symbol rows are obtained by folding $(q_\tau A^{n-\tau-\varkappa} \mid B_{\sigma-\varkappa} p_{n-\sigma})$, respectively $(q_{\tau-\alpha} B^{n-\sigma+\varkappa} \mid A_{\tau+\varkappa} p_{n-\sigma-\alpha})$ —according as $\sigma - \tau \gtrless 2\varkappa$ —with the ground form and the forms (41).

From the symbolic folding process one obtains the invariant differentiation processes in the familiar way by replacing the symbol rows $S^\sigma, T_{n-\tau}$ by the rows of differentiation symbols $\frac{\partial}{\partial p_\sigma}$ and $\frac{\partial}{\partial q_{n-\tau}}$. As usual, each product $\frac{\partial}{\partial p_{i_1 \dots i_\sigma}} \frac{\partial}{\partial q_{j_1 \dots j_{n-\tau}}}$ has the actual meaning $\frac{\partial^2}{\partial p_{i_1 \dots i_\sigma} \partial q_{j_1 \dots j_{n-\tau}}}$. Since the rows $\frac{\partial}{\partial p_\sigma}$ and $\frac{\partial}{\partial q_{n-\tau}}$ are cogredient with $S^\sigma, T_{n-\tau}$, they satisfy the same identities, in particular those between (42) and (42a), and between (43) and (43a). In summary:

Theorem VI. All invariant formations of arbitrary ground forms arise from them through repeated application of the symbolic folding process (41), or also of the invariant differentiation processes

$$\left(q_{n-\sigma-\lambda} \frac{\partial}{\partial p_\sigma} \mid \frac{\partial}{\partial q_{n-\tau}} p_{\tau-\lambda} \right), \quad (\sigma \geq \tau, \lambda = 1, 2, \dots, \min(\sigma, n - \tau)). \quad (45)$$

The original print and the current R823 source both give the paired upper endpoints $\sigma, n - \tau$ in (45), i.e. $\min(\sigma, n - \tau)$. The inherited English comparison witness changes them to $\tau, n - \sigma$. The source reading is retained here.

It remains to note that at every folding (41) the total weight of the form, that is, the sum of the weights of the individual variable rows (cf. § 1), decreases by the positive number $2(\lambda^2 + \lambda(\sigma - \tau))$. Independently of the general finiteness proof, this implies the finiteness of the form row, that is, of the totality of invariant formations of a given ground form that are linear in the coefficients.³

It should further be noted that Theorem VI remains valid for forms that do not satisfy conditions (12). Each factor in (3), $(S^\rho \mid p_\rho)^{m_\rho}$, can be replaced by a product of m_ρ linear factors $(A^\rho \mid p_\rho)(B^\rho \mid p_\rho) \cdots (M^\rho \mid p_\rho)$. The invariant formations thereby pass into those of a system of linear forms, which need only afterwards be set equal to one another. For every individual linear form, however, conditions (12), which contain only second-order differential quotients when differential symbols are introduced, are always satisfied.

²Clebsch, loc. cit., § 7.

³Cf. Clebsch, loc. cit., §§ 11 and 12: finiteness of the system of elementary covariants.